LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:15

Aim

To perform image transformation using the Direct Linear Transformation (DLT) method and obtain the transformation matrix using corresponding points.

Theory

The Direct Linear Transformation (DLT) is a method used to calculate a Homography matrix from corresponding points between two images.

In simple words:

Source points → DLT calculation → Homography matrix → Transformed image

At least 4 pairs of corresponding points are required.

Algorithm

1. Import OpenCV and NumPy.
2. Read the input image.
3. Select four source points from the image.
4. Define four destination points.
5. Construct the DLT equation using the corresponding points.
6. Use Singular Value Decomposition (SVD) to calculate the Homography matrix.
7. Normalize the Homography matrix.
8. Apply the matrix using cv2.warpPerspective().
9. Display the original and transformed images.
10. Press any key to close the output.

DLT Formula

For a source point (x, y) and destination point (X, Y):

The DLT method calculates the values of the 3×3 Homography matrix H.

CODE:

EXP-15.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

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```python
import cv2
import numpy as np

# Read image
image = cv2.imread("MICKEY.PNG")

if image is None:
    print("Error: Image not found!")
    exit()

height, width = image.shape[:2]

# Source points
src = np.float32([
    [100, 100],
    [width - 100, 100],
    [width - 100, height - 100],
    [100, height - 100]
])

# Destination points
dst = np.float32([
    [50, 50],
    [width - 50, 50],
    [width - 100, height - 50],
    [100, height - 50]
])


# -------- Direct Linear Transformation --------

A = []

for i in range(4):

    x, y = src[i]
    X, Y = dst[i]

    A.append([
        -x, -y, -1,
```

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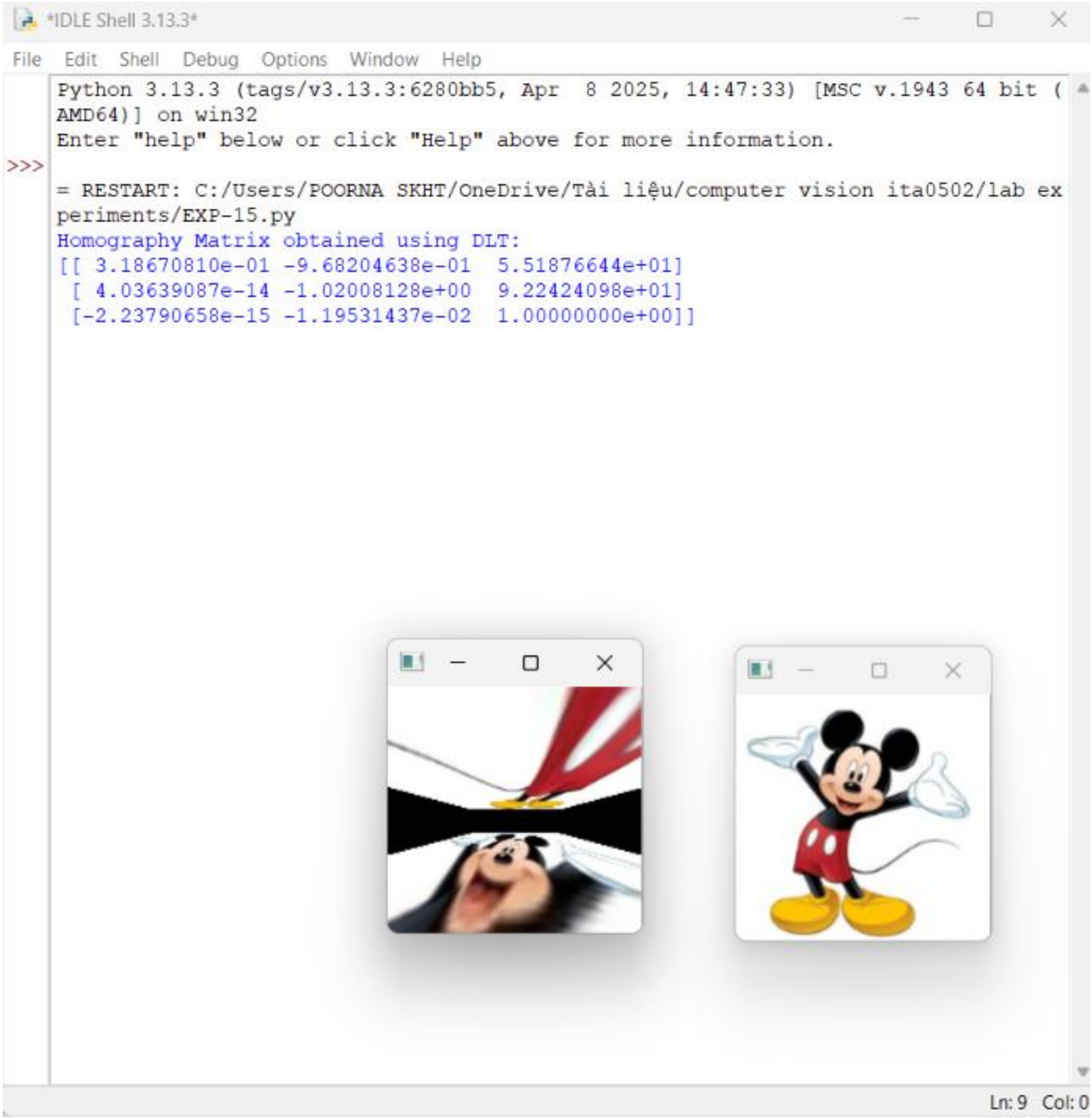
OUTPUT :



Result:

The Direct Linear Transformation method was successfully implemented to calculate the Homography matrix from four pairs of corresponding points. The calculated matrix was then used to perform perspective transformation on the image.